AI Tutor Bot – Finance Learning & Quiz Platform

Course Name: Gen AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: GAI-03

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Academic Year: 2022-26

Problem Statement & Objectives

1. Problem Statement

Traditional finance learning methods rely heavily on textbooks and static content, which often lack personalization, instant feedback, and self-evaluation mechanisms. Students face difficulty in understanding complex finance concepts and do not get immediate clarification or assessment of their learning progress. There is a need for an intelligent system that can provide interactive learning, quizzes, and evaluation one.

Traditional learning methods in finance education often face the following problems:

- Lack of personalized learning
- No instant feedback on answers
- Static content that does not adapt to learner level
- Difficulty in self-assessment

Students need a **smart tutoring system** that can explain concepts, generate quizzes, and evaluate answers instantly.

2. Project Objectives

The main objectives of this project are:

- To develop an AI-based tutoring system for finance learning
- To provide instant question-answer support using AI
- To generate quizzes automatically for concept evaluation
- To evaluate student answers and provide feedback
- To support structured and level-based learning

3. Scope of the Project

The scope of this project includes:

- Finance-focused AI tutoring
- Web-based learning platform
- Quiz generation and evaluation

Learning progress tracking

Out of Scope:

- Mobile application
- Payment system
- Multi-language support

Proposed Solution

The proposed solution is an AI Tutor Bot that provides:

The proposed solution is an **AI-powered Finance Tutor Bot** built using a Retrieval-Augmented Generation (RAG) architecture. The system combines a React-based frontend, a FastAPI backend, FAISS vector database, and Google Gemini AI to deliver accurate, source-backed finance learning. It enables students to ask questions, take quizzes, evaluate answers, and learn finance topics through structured lessons. The solution is scalable, modular, and designed for future cloud deployment.The system improves learning efficiency and reduces dependency on manual teaching.

Key Features

1. Instant question answering with explanations
2. Automatically generated finance quizzes
3. Answer evaluation with feedback
4. Structured learning modules based on student level

5. Overall Architecture / Workflow

The AI Tutor Bot follows a three-layer architecture consisting of frontend, backend, and AI processing layers. The user interacts with the web interface, which sends requests to the backend server. The backend communicates with the AI model to generate answers, quizzes, and learning content, and then returns the response to the user.

The system follows a three-layer architecture:

1. User interacts with the web interface
2. Backend processes requests and communicates with AI services
3. AI engine generates answers, quizzes, and learning content

Workflow:
User → Frontend → Backend → AI Engine → Backend → Frontend → User

User
↓
React Frontend(SPA)

Ask | Quiz | Evaluate | Tech
↓ REST API
FastAPI Backend
↓
RAG Engine

Prompt + Context
↓
FAISS Vector Database

(Finance Documents)
↓
Google Gemini AI

(Embeddings & Response)
↓
Response to User

(Answer / Quiz / Evaluation /

Learning Content)

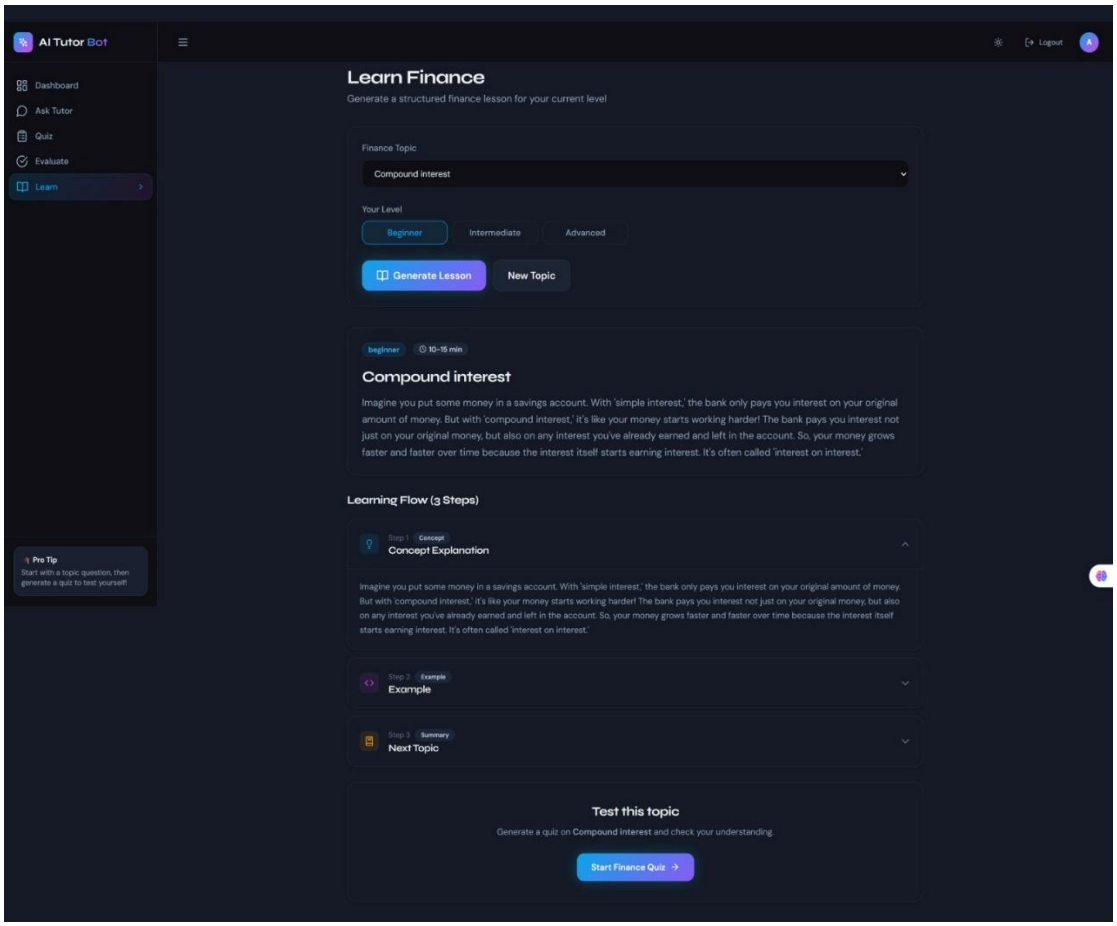
6. Tools & Technologies Used

Layer	Technology
Frontend	React.js (SPA)
Backend	FastAPI (Python)
AI Model	Google Gemini(LLM & Embeddings)
Retrieval Engine	FAISS Vector Database

Architecture	Retrieval-Augmented Generation (RAG)
API Communication	REST API
Data Storage	Local Files + Metadata JSON
Development Tool	VS Code
Deployment Mode	Localhost (Future Cloud Ready)

Results & Output

1. Screenshots / outputs



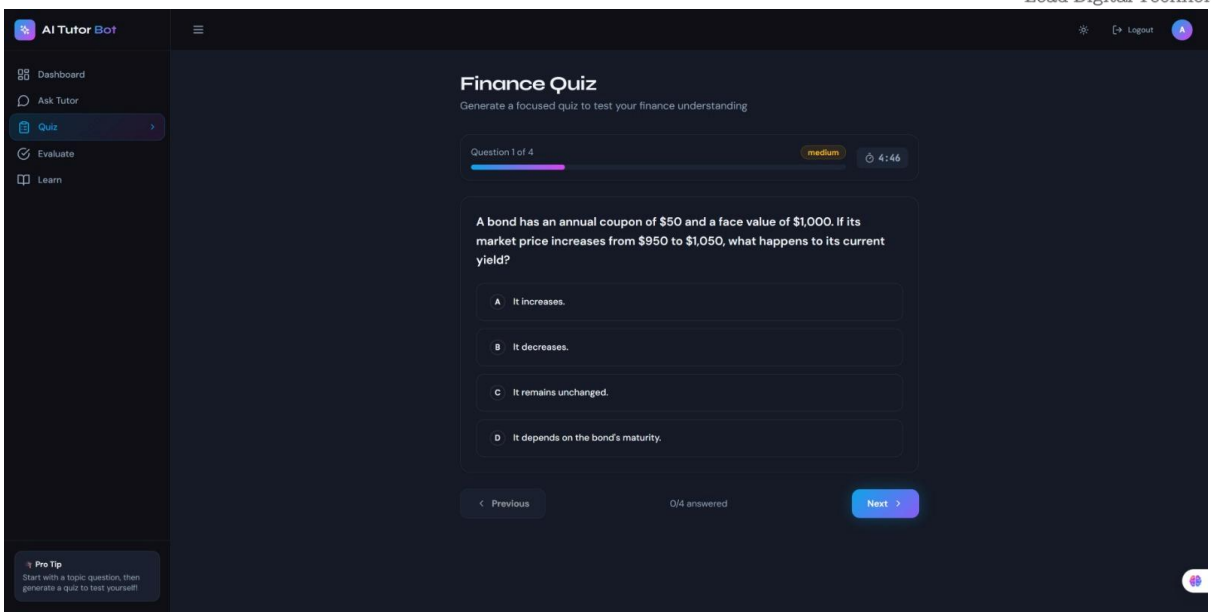


Figure 2: Dashboard interface of the AI Tutor Bot displaying learning progress, quiz attempts, average score, weekly activity, score trends, and recent activities.

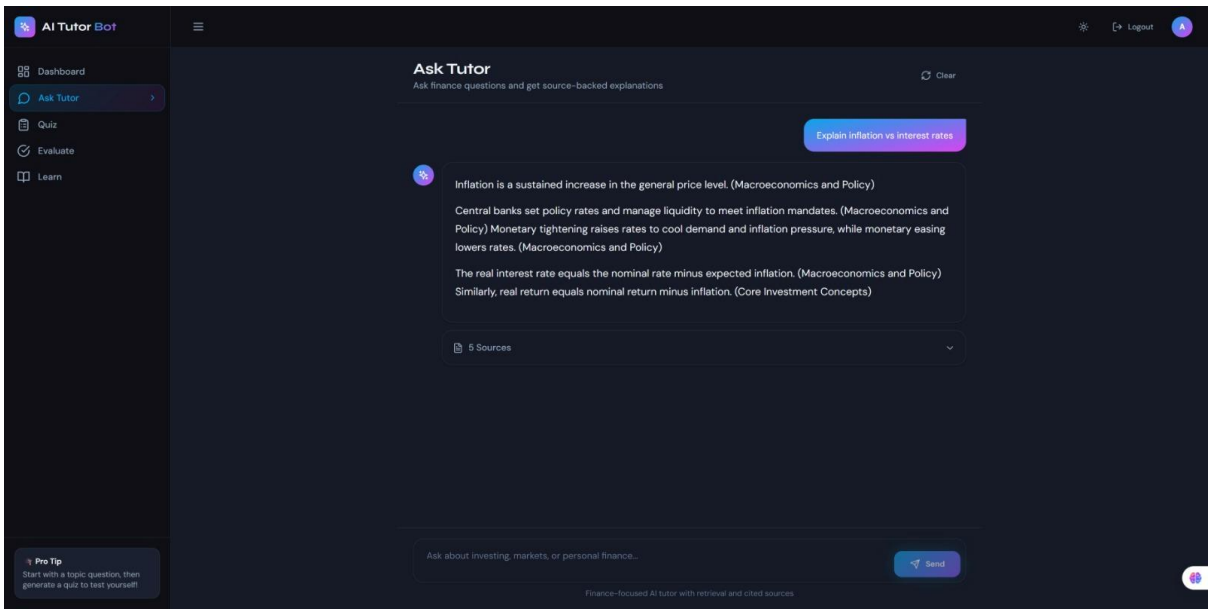


Figure 3: Ask Tutor module showing AI-generated, source-backed answers for finance-related questions entered by the user.

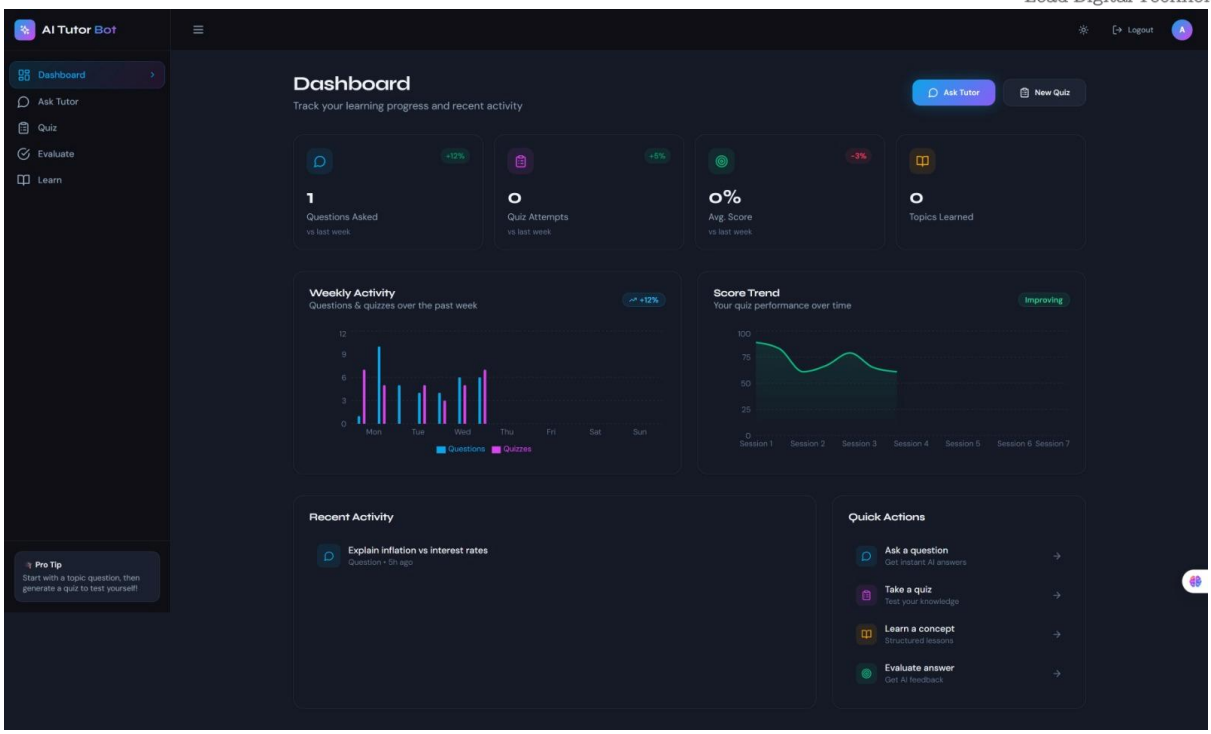


Figure 4: Finance Quiz module displaying AI-generated multiple-choice questions with difficulty level, timer, and navigation controls.

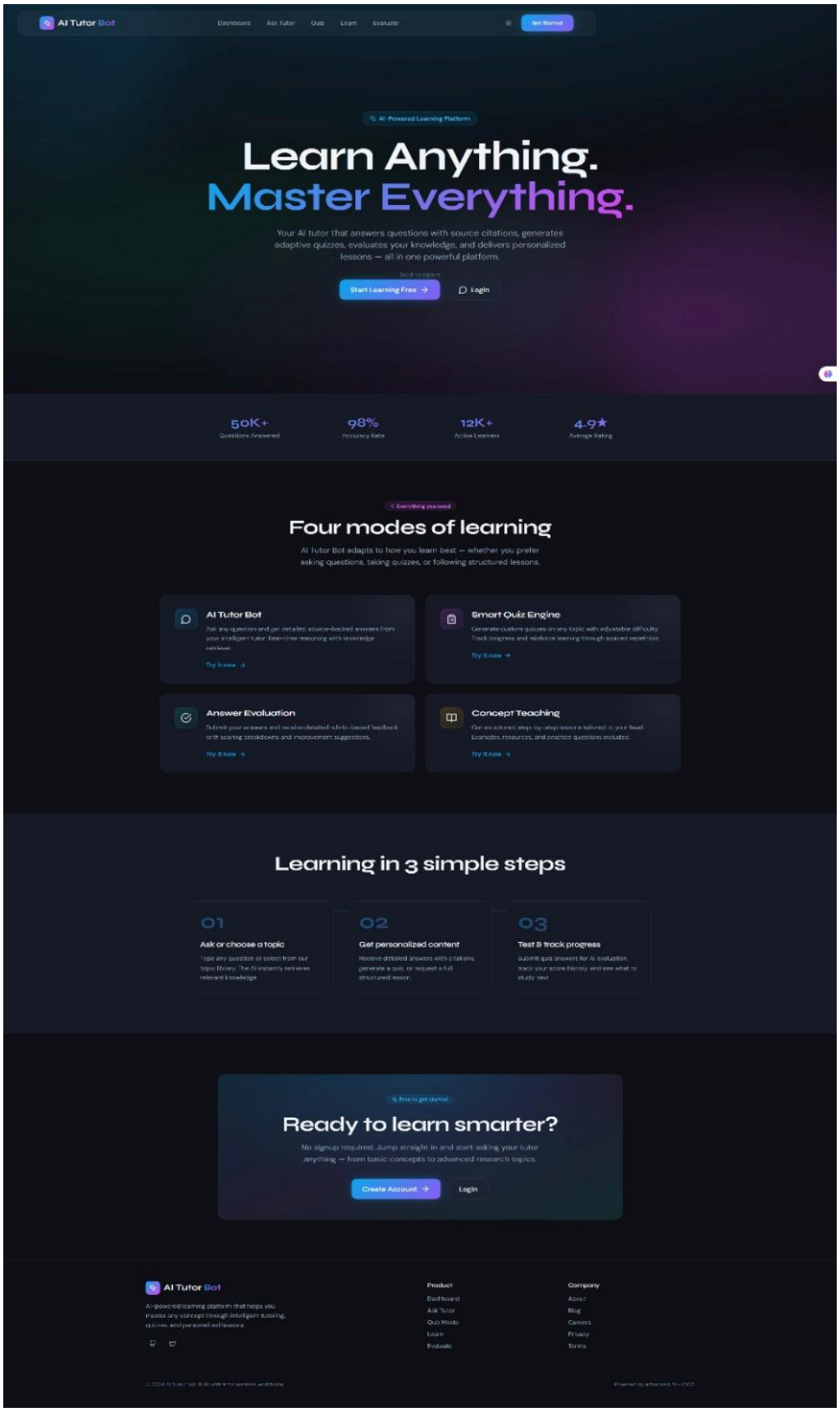


Figure 5 Learn Finance module showing topic selection, learning level selection, and structured lesson generation for finance concepts..

2. Reports / dashboards / models

In the current implementation, the project focuses on delivering an **AI-powered finance learning platform** using Retrieval-Augmented Generation (RAG). The system is designed mainly for **interactive tutoring, quiz generation, evaluation, and learning**, and does not include complex business intelligence dashboards or analytical reporting tools.

However, the system provides **learning and operational visibility** through the following components:

Reports

- Quiz Result Report: Displays quiz scores, correct answers, incorrect responses, and explanations generated by the AI model.
- Answer Evaluation Report: Provides structured feedback, scores, and reasoning for student answers evaluated using predefined rubrics.
- Learning Progress Report: Shows completed topics, learning level (beginner/intermediate/advanced), and recent learning activities.

Dashboards

- User Dashboard: Displays user activity such as number of questions asked, quizzes attempted, average scores, and recently learned topics.
- Learning Module Dashboard: Shows available finance topics, learning stages, and progress indicators for each topic.
- System Status View: Indicates backend availability and successful interaction between frontend, backend, FAISS vector store, and Google Gemini AI.

3. Key outcomes

The following outcomes were successfully achieved through this project:

- Successfully designed and implemented an **AI-powered Finance Tutor Bot** using a Retrieval-Augmented Generation (RAG) architecture.
- Developed an interactive web-based platform supporting **Ask, Quiz, Evaluate, and Teach** learning modes.
- Integrated **Google Gemini AI** with **FAISS vector database** to provide accurate, source-backed finance explanations.
- Implemented automatic **quiz generation and answer evaluation** to support self-assessment and personalized learning.
- Achieved a modular and scalable system design using **React frontend** and **FastAPI backend** as defined in the HLD.
- Improved learning effectiveness by providing **structured lessons, instant feedback, and progress tracking**.

Conclusion

The **AI Tutor Bot** project successfully demonstrates the use of **Artificial Intelligence in education**. The system provides an interactive, efficient, and personalized way to learn finance concepts.
This project enhanced our understanding of **AI integration, frontend-backend communication, and intelligent system design**, and reflects real-world application of modern learning technologies.

Future Scope & Enhancements

The following enhancements can be implemented in future versions:

- The following enhancements can be implemented in future:
- Support for multiple subjects
- User authentication & profiles
- Mobile application
- Advanced analytics dashboard
- Voice-based interaction
- Cloud deployment with scalability